

Supplement to “When Numerical Structure Reads the Recording Pipeline: A Boundary for Factorization-Based Data Analysis”

S1 Definitions, invariances, and the coordinate algorithm

L-profile. For $n = \prod_{i=1}^r p_i^{a_i}$ with distinct primes p_i , the log-shares are $\ell_j = a_{(j)} \ln p_{(j)} / \ln n$, sorted so $\ell_1 \geq \dots \geq \ell_r$, $\sum_j \ell_j = 1$. We read $L_1 = \ell_1$, $L_2 = \ell_2$, $L_3 = \ell_3$, $\text{Tail} = \sum_{j \geq 3} \ell_j = 1 - \ell_1 - \ell_2$, and $H_2 = \sum_j \ell_j^2$ (Simpson concentration of multiplicative energy). Per-record quantities are averaged over a channel’s records. The channel’s effective factor count is the ratio of averages $k_{\text{eff}} = (\sum_d w_d \mathbb{E}[H_2 \mid d]) / \overline{H_2}$, with w_d the channel’s digit histogram, $d(n) = \lfloor \log_{10} n \rfloor + 1$, and $\overline{H_2}$ the mean record-level concentration—this (not an average of per-record ratios) is what the analysis code computes, matching main-text Eq. (2). Records with fewer than three prime-power parts have $L_3 = 0$ structurally (see §S4).

What the profile retains of the primes. Every statistic of the sorted vector (ℓ_j) depends on n only through the multiset $\{a_i \ln p_i\}$: the profile discards prime *names* but retains prime *magnitudes* through the $\ln p_i$. It is therefore *not* invariant under prime relabeling—replacing a prime by one of a different size changes the profile ($6 = 2 \cdot 3$ has $L_1 = \ln 3 / \ln 6 \approx 0.613$; relabeling $3 \rightarrow 101$ gives $202 = 2 \cdot 101$ with $L_1 \approx 0.870$). This is precisely why small-prime composition can act as driver 3 of the main-text reduction: the smallness of the grid primes is visible to the profile. The accurate statement used throughout is that shape statistics are functionals of the log-share multiset $\{a_i \ln p_i\}$ —blind to prime names given their sizes.

Scale invariance of the residual coordinates. Multiplying a field by 10^c (a units change) adds c to both v_2 and v_5 of every record, where v_p is the p -adic valuation. The de-rounded coordinates are computed on the $2 \cdot 5$ -free core $n / (2^{v_2} 5^{v_5})$, which is unchanged by the 10^c factor: $2^{v_2(10^c n)} 5^{v_5(10^c n)} = 10^c \cdot 2^{v_2(n)} 5^{v_5(n)}$. All scale-clean statistics are therefore exactly invariant under decimal unit scaling. This invariance is what exposes the “domain signal” of the main text §4 as a units artifact: it reverses sign on monetary channels once the coordinates are made scale-clean (the de-rounded second log-share residual: raw $dL_2 + 0.042 \rightarrow$ scale-clean $dL'_2 - 0.045$). The grid mass and grid depth are by design *not* scale-invariant—they are the units/grid readout—and are corrected against the magnitude-matched null.

Algorithm S1 Recording-grid coordinates of a channel (eleven values)

Require: integers $n_1, \dots, n_N$; modal grid g_0 ; moduli $\mathcal{M} = \{100, 25, 49, 47, 43\}$

```
1:  $\rho, \tau, \gamma, c_2, c_5 \leftarrow 0$ ; counts  $C_m[\cdot] \leftarrow 0$  for  $m \in \mathcal{M}$ 
2: for  $k = 1$  to  $N$  do
3:    $a_2 \leftarrow v_2(n_k)$ ;  $a_5 \leftarrow v_5(n_k)$ ;  $L \leftarrow \ln n_k$ 
4:    $c_2 \leftarrow c_2 + a_2 \ln 2 / L$ ;  $c_5 \leftarrow c_5 + a_5 \ln 5 / L$ ;  $\tau \leftarrow \min(a_2, a_5)$ 
5:   if  $g_0 \mid n_k$  then  $\gamma \leftarrow \gamma + 1$ 
6:   for  $m \in \mathcal{M}$  do  $C_m[n_k \bmod m] \leftarrow C_m[n_k \bmod m] + 1$ 
7:   end for
8: end for
9:  $\rho \leftarrow (c_2 + c_5) / N$ ; normalize  $\tau, \gamma, c_2, c_5$  by  $N$ 
10: for  $m \in \mathcal{M}$  do  $Q_m \leftarrow \frac{1}{2} \sum_r |C_m[r] / N - 1/m|$ 
11: end for
12:  $\bar{d} \leftarrow \text{median}_k d(n_k)$ 
13:  $\rho_{\text{lo}}, \rho_{\text{hi}} \leftarrow \text{mean } \rho(n_k) \text{ over } d(n_k) \leq \bar{d} \text{ and } > \bar{d}$ 
14: return  $(\rho, \tau, \gamma, c_2, c_5, \{Q_m\}, |\rho_{\text{lo}} - \rho_{\text{hi}}|)$ 
```

Recording-grid coordinates (main text §3), full set. Grid mass $\rho(n) = (v_2(n) \ln 2 + v_5(n) \ln 5) / \ln n$ with its c_2/c_5 additive split; grid conformance $\gamma = \frac{1}{N} \sum_k \mathbf{1}[g_0 \mid n_k]$ for the channel’s modal grid g_0 ; grid depth $\tau(n) = \min(v_2(n), v_5(n)) = \max\{t : 10^t \mid n\}$; residue quantization $Q_m = \frac{1}{2} \sum_{r=0}^{m-1} |\widehat{\text{Pr}}(n \equiv r \bmod m) - \frac{1}{m}|$ at $m \in \{100, 25, 49, 47, 43\}$, each reported against a magnitude-matched uniform-null expectation; and the digit-conditioned grid-mass split $|\rho_{\text{lo}} - \rho_{\text{hi}}|$ across the channel’s median-digit split (a mixed-era flag). Algorithm S1 computes all eleven channel-level coordinates in $O(N \log n_{\max})$ from stored integers alone.

S2 Null construction and the core-magnitude baseline

Magnitude-conditioned baseline. The reference tables $\mathbb{E}[\cdot \mid d]$ are built from 2×10^5 seeded uniform draws per digit stratum $d = 1..18$ (frozen seed 20260702), factorized exactly. $\mathbb{E}[L_1 \mid d]$ falls from 0.952 ($d=1$) to 0.633 ($d=18$); the Poisson–Dirichlet limit value is 0.6243 (Dickman 1930; Billingsley 1972; Kingman 1975; Arratia et al. 2000). Derived baselines exist for the de-rounded core (seed 20260703) and the deep-stripped core (seed 20260706). Channel residuals weight the stratum expectations by the channel’s own digit histogram.

Core-magnitude re-indexing (the reference correction). Stripping primes $\leq c$ from n yields the deep core $\kappa_c(n) = \prod_{p_i > c} p_i^{a_i}$. The naive reference compares the core’s profile to $\mathbb{E}[\cdot \mid d(n)]$, indexed by the *original* digit length. But rounded values carry more small-prime mass, lose more magnitude when stripped, and have smaller cores—and smaller integers are intrinsically more concentrated. The correct reference indexes by the core’s own magnitude: draw uniform integers over the original strata, strip identically, tabulate $\mathbb{E}[\text{Tail} \mid d_\kappa]$ against the core digit length d_κ , and weight by the channel’s normalized core digit

Algorithm S2 Core-magnitude re-indexing

Require: channel values $\{n_k\}$; strip cut c (here 7); core-magnitude table $\mathbb{E}[\cdot \mid d_\kappa]$

- 1: build the table once: uniform draws over strata 1..18, strip by κ_c , tabulate vs d_κ
 - 2: **for** each n_k **do**: $\kappa \leftarrow \kappa_c(n_k)$; **if** $\kappa = 1$ **skip**; accumulate $\text{Tail}_c \equiv \text{Tail}(\kappa)$, $H_2(\kappa)$; bump $w[d_\kappa]$
 - 3: **end for**
 - 4: **return** $d\text{Tail}^{\text{core}} = \overline{\text{Tail}_c} - \sum_{d_\kappa} \frac{w[d_\kappa]}{\sum w} \mathbb{E}[\text{Tail} \mid d_\kappa]$ and $k_{\text{eff}}^{\text{core}}$
-

histogram (Algorithm S2). Under this reference the surviving arm-domain concentration excess collapses from -0.0184 to -0.0001 (90% family-bootstrap CI $[-0.0004, +0.0004]$); the exponent-stacking mechanism check (1.9% of observational records versus 100% of genuine smooth products) confirms no real concentration mechanism was present.

Local-flatness stress (v10.3, exploratory; seed 20260718). A simulation maps where the model’s local-flatness assumption fails (`stage25_flatness.py`, plan frozen in `run_config_flat25.json`; output `flat25_results.csv`). Latent draws $x \sim \text{Exp}(\text{scale} = r g)$ recorded on grids $g \in \{10, 100\}$: at curvature-to-grid ratio $r = 0.5$ the recorded quotient is far from generic (dL_1 of the quotient -0.18 , quotient k_{eff} 1.34, decimal-prime log-share 0.64 against a 0.36 generic reference); deviations shrink monotonically to $|dL_1| \leq 0.013$ and k_{eff} within 0.02 of 1 for $r \geq 5$. A zero-inflated count case (Poisson $\lambda = 30$, 60% zeros, $g = 10$) shows the same small-support bias ($dL_1 + 0.048$, k_{eff} 0.94). Under the frozen 2 SE adequacy rule (severe at $N = 4 \times 10^4$: 2 SE ≈ 0.002) only the $r = 5$, $g = 100$ cell passes exactly; the practical reading is the magnitude ordering above. Direction of the induced bias: a *false* composite/grid signature (inflated k_{eff} and decimal-prime mass), never a masked process signal. Named failure regimes: narrow-support latents ($r \lesssim 2$), sparse zero-inflated counts on coarse grids, and benchmarked estimates (the W12-1 mechanism of §S6).

S3 Corpus construction and quarantine

A *channel* is one integer-valued column of one dataset: $\geq 99\%$ of non-null entries integers, ≥ 100 records, maximum ≥ 2 , deduplicated by content fingerprint. Integerization is canonical: absolute values; missing and non-integer entries dropped; $|n| \leq 1$ excluded (no factorization); monetary fields scaled to integer cents (round-half-to-even); values above 10^{18} logged and skipped. Channels group into dataset-families (one source lineage) and domains (named process areas). A prospectively specified quarantine reduced a provisional map of 462 candidate families in 15 domains to the frozen corpus of 18,951 channels, 367 families, 18 domains (the quarantine’s relabeling re-partitioned the 15 provisional domain labels into the 18 frozen ones) ($\sim 9.6 \times 10^8$ exact factorizations), reassigning mislabeled generator families; the frozen manifest records the patch and hashes. Two properties are load-bearing for scope: a few high-volume administrative sources dominate channel counts (network telemetry, procurement, and census are $> 90\%$ of channels), and domain labels are heterogeneous (18) though uneven. The deviation log (16 entries) records data losses and access changes (e.g., deletion of unpacked archive snapshots; API key requirements).

Raw-map teardown numbers (main text §4). Three-cluster structure in the family-level (L_1 , Tail) plane: silhouette 0.719; cluster survival after dividing out $2 \cdot 5$: Jaccard 0.39–0.43; scale-clean sign reversal of the monetary domain signal ($+0.042 \rightarrow -0.045$); intrinsic dimension of the cleaned space 1.16 (TwoNN); residue-quantization negative controls: planted resonant families score 0.051–0.167 against planted generic ≈ 0.0056 (~ 9 – $30\times$), observed channels median 0.020, 95th percentile 0.439.

Leave-one-source-out robustness (v10.3, exploratory; seed 20260716). The four-driver arm sequence and the core-magnitude collapse were rerun excluding network telemetry, procurement, or census wholesale, and with equal-family weighting (`stage23_loso.py`, holds-rule frozen in `run_config_loso23.json`; outputs `loso23_results.csv`, `loso23_family_counts.csv`). Telemetry and census contribute no families to the frozen arm-domain set (equity markets, seismology, real estate, procurement, food_nutrition), so their exclusion leaves the arm untouched by construction—the residual claim never rested on them; procurement’s exclusion removes 8 of 166 arm families. Every variant satisfies the frozen rule: raw arm Tail excess $+0.091$ to $+0.092$; de-rounded -0.021 ; deep-strip -0.018 to -0.019 ; core-indexed -0.0001 to -0.0002 with all 90% family-bootstrap CIs containing 0 (telemetry-excluded: -0.0001 [-0.0004 , $+0.0004$]; procurement-excluded: -0.0002 [-0.0005 , $+0.0003$]). Per-variant per-domain family counts accompany the output table.

S4 The L1/L3 geography disposition (Build geo15)

This exploratory build (seed 20260715) resolves the previously unresolved question of what the L1/L3 Cartesian geography means. Its analysis plan and disposition rule were frozen in `run_config_geo15.json` before results were computed. No frozen confirmatory rule covers it; the $\Delta\text{AUC} \approx 0.05$ retention threshold is a project-management rule. Full narrative: `L1_L3_DISPOSITION.md`.

Observational level. Per-record L-profiles were never persisted by the frozen pipeline; the finest stored grain is the channel. Analyses run at three levels: per-record (stratified recompute: 153 channels, 17 domains, 9,199 per-record profiles), channel (18,992 stored channel profiles), and family (462; the blocking unit).¹ 37.5% of sampled records have fewer than three prime-power parts, so per-record L_3 is a structural zero for over a third of records (conditional mean when defined: 0.158); channels dominated by such records (mean parts < 2.5) are faceted separately in every display.

¹Three related counts appear in this paper and are defined here once. The frozen post-quarantine corpus has 18,951 channels in 367 families (the main-text headline). The stored channel-profile table was computed on the pre-quarantine observational set of 18,992 channels in 408 families: the frozen 18,951/367 plus the 41 quarantined generator channels (one family each, all carrying the `unresolved` domain label; these are distinct from the 159 in-corpus `unresolved` channels of main-text Table 1), which every geo15 model excludes via the `unresolved/synthetic-control` filters. The 462 figure is the cardinality of the full frozen family label map (observational plus synthetic-control families); grouped holdout blocks on the `dataset_family` column of that map, so no two channels of one family ever cross a fold boundary, and after the exclusions the models see the frozen observational families only. Main-text Figure 6 summarizes family-level k_{eff} over the 367 frozen-corpus families.

Table S1: Held-out ladders (exploratory; domain task over the 12 domains with ≥ 5 families; macro OVR AUC, family-grouped bootstrap 90% CI). Source: `geo15_heldout_auc.csv`.

feature set	domain AUC	90% CI	Δ vs R	Δ vs R+ L_1
R (recording variables)	0.512	[0.497, 0.528]	—	—
R + L_1	0.522	[0.512, 0.531]	+0.009	—
R + L_1 + L_3	0.535	[0.514, 0.554]	+0.022	+0.013
R + full profile	0.547	[0.525, 0.566]	+0.035	+0.025
count vs amount (binary) R alone 0.9999; adding L_1, L_3 changes nothing				
Deltas before rounding: +0.0091, +0.0222, +0.0131, +0.0349, +0.0258. Paired family-bootstrap 90% CIs for the deltas are given in the text below.				

Block A: raw geography. Recomputed uniform-integer pseudo-channels match the frozen baseline within 0.02 at every stratum (machinery validation). Exact products of 2/3/4 primes sit at $(L_1, L_3) = (0.51, 0.00)/(0.35, 0.32)/(0.28, 0.25)$; additive controls at the baseline $(0.67, 0.08)$; the same exact products after one decimal rounding (grid 10/100/1000) land at L_1 0.47–0.60, L_3 0.10–0.15—inside the recorded band.

Block B: controlled geography. Control sequence at channel level (domains with ≥ 5 families; synthetic and unresolved channels excluded): raw $(L_1, L_3) \rightarrow$ digit-length residuals $\rightarrow$ de-rounded ($2 \cdot 5$ -stripped) residuals $\rightarrow$ residue-conditioned residuals. Out-of-family domain separability of the 2-D coordinates (GroupKFold by family, macro OVR AUC, family-bootstrap 90% CI): 0.526 [0.508, 0.543] raw; 0.422 [0.298, 0.527]; 0.513 [0.495, 0.526]; 0.522 [0.509, 0.531]. Family-level silhouettes by domain are negative at every stage (no domain islands). The recomputed digit-length residual dL_1 matches the frozen stored value to 10^{-5} .

Block C: held-out incremental information. Grouped holdout (GroupKFold, 5 folds, groups = dataset-family; never record-level CV), gradient-boosting classifier, family-grouped bootstrap (500) CIs. Table S1. L_3 beyond recording variables + L_1 : a small positive increment of +0.013, below the pre-declared 0.05 retention threshold (a project rule); the binary count-vs-amount task is at 0.9999 from recording variables alone. Two calibration caveats are stated plainly. First, paired family-bootstrap 90% CIs (stage29, seed 20260722; one family resample shared by both ladders per draw) quantify the deltas at the family grain: L_1 beyond R +0.009 [−0.006, +0.026]; L_3 beyond R+ L_1 +0.013 [−0.011, +0.038]—the interval includes 0; full profile beyond R +0.035 [+0.004, +0.073], positive but below the threshold. Second, the incremental- Δ AUC ladder is *uncalibrated for hidden-process signal*: the task-sensitivity control of this section calibrates the design against *recording* differences only, so a small Δ AUC bounds what these coordinates add to this task, not what any conceivable process signal could produce. Consequence (disposition rule 4): the manuscript makes no claim that L_1/L_3 topology is independently informative.

Task-sensitivity positive control (v10.3, exploratory; seed 20260717). The transfer design itself is calibrated by a positive control (`stage24_taskpos.py`, plan frozen in `run_config_taskpos24.json`; outputs `taskpos24_calibration.csv`, `taskpos24_permutation.csv`, `taskpos24_sl_diagnostics.csv`). Synthetic pseudo-channels (150 families $\times$ 3 channels $\times$ 600 records) with class labels defined by injected recording differences run through the *identical* family-grouped pipeline: AUC rises from 0.47–0.49 at zero effect to 0.67/0.81 (raw/digit-residual coordinates) at a grid-mass shift of +0.020, 0.99 at +0.066, and 1.00 at full-coverage grids; a tick-7 treatment with near-zero grid-mass shift is read at 1.00 through the shape coordinates. A family-level label-permutation null centers on 0.485 (range [0.39, 0.61] over 20 draws; one draw marginally exceeded the frozen 0.6 bound and is reported as such). The design is therefore sensitive to recording differences an order of magnitude smaller than what would be needed to explain the real corpus’s transfer AUC of 0.526 as a design failure. Diagnostics of the digit-residual stage’s below-chance point (0.422, wide CI): GroupKFold with the corpus’s extreme family-size skew yields one fold with a single test family and class (macro-AUC undefined), one fold 98% dominated by a single class (0.864), and three balanced folds at 0.497–0.509—fold instability at near-zero signal, not negative information.

Block D: mechanism stress. Per-record recompute under transforms on the same subsample. A three-factor exact product moves from (0.35, 0.32) to (0.62, 0.10) under $n \rightarrow n+1$ (the 12-digit baseline point is (0.64, 0.09)); one grid rounding does the same. Recorded channels move most when the grid itself is removed (mean $|\Delta L_1|$: +1 shift 0.078; de-round 0.159; strip ≤ 7 0.238)—the grid is where the shape mass lives. Subsample domain separability from (L_1, L_3) : identity 0.624, after $n+1$ 0.521.

S5 Spike-in machinery, tiers, and full results

Hosts: the six amount-kind channels (capped at 15,000 records); eligible records have $2 \cdot 5$ -free core $\kappa \geq 300$. With $t_i = \log_{10} \kappa_i$ the host record’s core magnitude, the injected product Q_i comes from three tiers: (a) *naive*—balanced two-factor products at the host’s marginal core-magnitude distribution, placed on the modal grid at random positions; (b) *magnitude-matched*—in place: $t = t_i$ and the record’s exact grid part is reused, so the mixture’s (grid, magnitude) joint is host-exact; (c) *base-matched adversarial*— $Q_i = 3^{b_3} 7^{b_7} R$ with (b_3, b_7) drawn from the host’s empirical deep-core joint and R a balanced two-factor product coprime to $210 = 2 \cdot 3 \cdot 5 \cdot 7$, matching grid, magnitude, *and* the small-prime joint, leaving composite structure of the > 7 core as the only difference from a generic quotient.

Detection: a channel is recovered at mixture f iff the replicate-median $k_{\text{eff}}^{\text{core}}$ (15 replicates) exceeds the channel’s own $f=0$ record-bootstrap 95th percentile ($R = 400$, n -matched). Confound check: the eleven grid coordinates of the $f = 0.10$ mixture are z -scored against 200 host bootstraps; a tier passes iff $\max |z| < 3$.

The earlier off-grid design (exact products into rounded hosts) drifts the mixture’s grid coordinates at $|z|$ up to 10^7 and is retained only as an illustration of why the confound check is mandatory.

Table S2: Quotient-level spike-in recovery (fraction of six host channels) by tier and mixture fraction, with the mixture-level confound check. Source: `power13_quotient_spikein.csv`, `power13_confound_check.csv`.

injection tier	$f=0.01$	$f=0.05$	$f=0.10$	$f=0.25$	$\max z $ (med/max)	clean hosts
(a) naive	0.50	1.00	1.00	1.00	15.8 / 21.5	0/6
(b) magnitude-matched	0.50	1.00	1.00	1.00	1.8 / 6.5	5/6
(c) base-matched adversarial	0.00	1.00	1.00	1.00	2.0 / 2.9	6/6

Count-kind hosts (v10.3, exploratory; seed 20260719). The tier-(b) in-place design was extended to count-kind hosts (`stage26_counthosts.py`, host recipe and holds-rule frozen in `run_config_count26.json`; outputs `count26_*.csv`). The frozen magnitude screen (median $\geq 10^5$, ≥ 500 eligible cores) selects three telemetry byte-count channels from distinct families—the only count-kind channels at the required magnitude, which itself scopes the claim. Recovery: 0/3 at $f = 0.01$, 3/3 at $f = 0.05$ and above (replicate-median $k_{\text{eff}}^{\text{core}}$ against each channel’s own $f=0$ bootstrap 95th percentile, $R = 200$). The mixture-level confound check exceeds the $|z| < 3$ bar in all three hosts (max $|z|$ 3.3–5.2), so per the frozen rule the extension does *not* qualify as clean recovery; the per-coordinate decomposition, however, shows every flagged coordinate is a *residue* score drifting *negative* (tv_{100} , tv_{25} , tv_{47} , tv_{43}) with the decimal-grid coordinates (grid mass, depth, conformance, c_2 , c_5) within noise: the injections dilute the hosts’ strong intrinsic resonance, the same mechanism as the one flagged amount-kind host, not grid leakage. Both the recovery and the flags are reported, and Section 7 of the main text is scoped accordingly.

Power curves and false-positive calibration (v10.3; seed 20260721). Assembled from the frozen power13 outputs without recomputation (`stage28_powerfp.py`; outputs `powerfp28_curves.csv`, `powerfp28_fpcal.csv`): per-tier recovery is 0.5, 1.0, 1.0, 1.0 (naive and magnitude-matched) and 0.0, 1.0, 1.0, 1.0 (adversarial) of six hosts at $f = 0.01, 0.05, 0.10, 0.25$, with per-host effect sizes in units of each channel’s own noise band. False-positive calibration is by construction of the record-bootstrap band (nominal 0.05); the single frozen $f=0$ observation per host sits below its threshold in 6/6 hosts, and the count-host extension uses the same construction.

S6 Pre-registration ledger and misfire postmortems

Epistemic status of the registrations. The decision rules were frozen in configuration files inside the analysis tree (hash-integrity via the artifact manifests), not lodged with an external registry, so they are *internally frozen* rather than externally timestamped; readers should weight them accordingly. Two further distinctions matter. First, the acquisition-frozen runs (emp11, whisp12, vote14) wrote their rules before any network fetch, so those rules were evaluated on data the authors had never seen; the computation-frozen runs (gate07, hunt08, hunt09, hard10, power13) wrote their rules before the computation but over corpus data that existed locally during rule development. Second, seed values follow a date-like labeling convention (20260702, 20260707, ...); they are labels, not timestamps,

Table S3: All twenty frozen decision rules with outcomes.
 artifact/manifests/decision_rules.csv.

Source:

run	id	rule (frozen) → outcome	verdict
gate07	G1	product AUC $\geq 0.85 \rightarrow 0.86\text{--}0.96$	met
gate07	G2	one digit of rounding destroys detection → chance at grid 100	met
hunt08	H8-1	structured products AUC $\geq 0.85 \rightarrow 0.889\text{--}0.998$	met
hunt08	H8-2	prime-pair $k_{\text{eff}} \approx 1.1 \rightarrow 1.035$	met [†]
hunt09	H9-1	residual $\rightarrow 0$ under core-magnitude null $\rightarrow -0.0184 \rightarrow -0.0001$	met
hunt09	H9-2	COD multiplicity $k_{\text{eff}} < 1 \rightarrow 0.534$	met
hunt09	H9-3	NIST $k_{\text{eff}} \neq 1$ (direction uncertain) $\rightarrow 1.707$	met
hard10	H10-1	semiprime: k_{eff} blind, full-profile sees $\rightarrow 0.44/0.93$	met
hard10	H10-2	off-grid spike-in recovered $\rightarrow 6/6$	met
hard10	H10-3	$n \rightarrow n+1$ collapses detection $\rightarrow \text{AUC } 0.44$	met
hard10	H10-4	grid coordinates beat Benford on grid mismatch $\rightarrow 1.00$ vs 0.46	met
emp11	E11-1	employment multiplicative null $\rightarrow k_{\text{eff}} 0.98\text{--}1.00$	met
emp11	E11-2	survey grid-marked vs census $\rightarrow \rho 0.474$ vs 0.102	met
whisp12	W12-1	replicate-or-retire whisper $\rightarrow$ no CI-clean replication	missed
power13	P13-1	quotient spike-in powered $\rightarrow 6/6$ (adversarial clean)	met
power13	P13-2	energy omnibus recovers pairs $\rightarrow$ underpowered (0.17)	missed
power13	P13-3	atom count decreases $\rightarrow$ direction inverted	missed
power13	P13-4	grid coordinates add value on subtle grid $\rightarrow 1.00$ vs ≤ 0.53	met
vote14	V14-1	votes multiplicative null $\rightarrow k_{\text{eff}} 0.98\text{--}1.03$	met
vote14	V14-2	votes vs modeled CVAP grid-marked $\rightarrow \rho 0.275$ vs 0.12	met

[†]The H8-2 expectation was stated qualitatively in the frozen text (“ ≈ 1.1 ”); the recorded verdict applied the run’s own criterion (elevated above 1 with detection AUC near chance), which 1.035 met. No quantitative tolerance was frozen; we state this rather than imply one.

and carry no registration force of their own (§S9). The v10.3 exploratory extensions follow the same internal-freeze protocol and are labeled exploratory throughout.

P13-2, the underpowered omnibus. Energy distance to a fixed generic reference cloud was pre-registered as the primary statistic for balanced-pair injections ($\geq 4/6$ channels at $f \leq 0.10$); it recovered at most 0.17. Cause: dilution—a real host already sits far from a generic cloud, so a small injected fraction barely moves the group statistic. The targeted stripped-core statistics recover 6/6 at $f = 0.05$. Implication: power lives in targeted core statistics keyed to the specific alternative; a fixed reference cloud is the wrong null for an off-cloud channel.

P13-3, the inverted direction. The mean atom count of the stripped core was pre-registered to decrease under balanced-pair injection (intuition transferred from a synthetic null with many-part comparators). On real hosts the direction inverts: real cores average ≈ 1.1 atoms, so two-atom injections *raise* the count. Correct arithmetic against the wrong baseline; the conclusion rests on $k_{\text{eff}}^{\text{core}}$ and tail statistics, whose directions held. Implication: coordinate directions must be calibrated on the real host baseline.

W12-1, the vanishing whisper. A candidate multiplicative residual in preliminary model-based survey estimates (scale-clean first log-share residual $dL'_1 = +0.016$) was registered for promotion only if it replicated CI-clean, same sign, effect ≥ 0.01 , in an independent

estimation system. Across four systems (CES real-time vintages, PEP, seasonal adjustment, ACS) the sign was unstable and no contrast was CI-clean; the signal existed only in the preliminary vintage and vanished after benchmarking—itsself an instance of the recording heuristic: benchmarking is a later additive re-anchoring that severs multiplicative structure. The rule correctly retired the candidate; sub-0.02 multiplicative residuals in survey estimates are estimation-model transients outside the instrument’s envelope.

S7 Exact-integer catalogue and computation

Catalogue. Synthetic ground truth (magnitude- and moment-matched): sums and Poisson tallies $k_{\text{eff}} = 1.00$ exactly; products of k factors AUC 0.86–0.96; detection survives factor imbalance to a 90/10 log-split and is destroyed by one digit of rounding. Structured products with closed-form factorizations (magnitudes to 10^{7263}): factorials, binomials, multinomials, primorials, B -smooth numbers, and finite-group orders (e.g. $|GL(k, q)| = q^{k(k-1)/2} \prod_{i=1}^k (q^i - 1)$) all clear the pre-registered AUC ≥ 0.85 bar (0.889–0.998). Measured fields: crystallographic site multiplicities (Gražulis et al. 2024) $k_{\text{eff}} = 0.534 \pm 0.091$ ($n = 11,240$), atomic degeneracy products (NIST 2024) 1.707 ± 0.557 ($n = 505$), graph automorphism orders (Leskovec and Krevl 2014) exact to $10^{17,265}$ with k_{eff} up to 4.44; real additive fields 0.996. Balanced prime pairs are the edge case: k_{eff} -blind asymptotically (for distinct primes p, q , $H_2 = \frac{1}{2} + 2(\ell_1 - \frac{1}{2})^2 > \frac{1}{2}$ strictly, meeting the null only in the magnitude limit) yet recoverable from full-profile statistics at AUC 0.83–0.93.

Exact computation. No value above the sieve range is factorized by trial division, and no profile is computed through floating point. Exponent maps come in closed form: Legendre’s $v_p(n!) = \sum_{k \geq 1} \lfloor n/p^k \rfloor$ for factorials (Legendre 1808); Kummer’s carry count for binomials/multinomials (Kummer 1852); closed factorization of group orders; twin-class reduction plus Schreier–Sims for automorphism orders (Seress 2003). Arbitrary-precision arithmetic certifies residual large factors; the only floating-point step is $\ln n = \sum_i a_i \ln p_i$ from the exact exponent map. There is no path by which float overflow could manufacture the reported k_{eff} values.

S8 Benchmarks: digit battery versus grid coordinates

(Dashes in the Beber–Scacco column are N/A by design: the test is defined for election-return-style count series and was run only on the contrasts involving them.) Two readings of the fabrication row, both reported: under the frozen one-sided orientation the grid coordinates “add value” at $f = 0.25$ because the one-sided digit battery anti-detects; read two-sided, a last-two-digit drift test reaches AUC 1.00 and dominates. On real returns the grid coordinates add detection value nowhere; their detection edge lives on non-decimal grids (constructed case), and their real-data role is attribution.

Table S4: Attribution vs. detection: naive digit battery (incl. the Beber–Scacco vote-forensic test) vs. grid coordinates, by contrast (AUC; ties retained). Source: `power13_naive_benchmark.csv`, `vote14_benchmark.csv`.

contrast	terminal- digit bat- tery	Beber– Scacco	grid coords	role
survey vs. census, employ- ment (decimal grid)	1.00	—	1.00 (tie)	calibration
ACS point estimates vs. census	0.29–0.54	—	0.77	below bar
ACS margin-of-error vs. census	0.74–1.00	—	1.00 (tie)	attribution
enumerated votes vs. modeled CVAP (5-grid)	1.00	fires on CVAP	1.00 (tie)	attribution
real-return fabrication contrast (2-sided)	1.00	anti- detects	0.68–0.83	detection lost
constructed 47-tick vs. 43-tick	≤ 0.53	—	1.00	detection
era-mixture fabrication	0.46–0.74	—	0.49 (both fail)	neither

S9 Artifact, manifests, and reproduction

The deposit contains: the 21 frozen v8-era configuration files (`run_config_*`) plus the exploratory geo15 and six v10.3 configs; the decision-rule manifest (20 rules); `claim_manifest.csv` (36 claims, each mapped to script, output table/figure, config, and SHA-256); the deviation log (16 entries); all analysis scripts (stages 1–21, plus the exploratory `stage22_geo15.py` with unit tests) and figure scripts (`scripts_v10/`); hash manifests for tables and figures; and a worked end-to-end verification of the CVAP χ^2 claim in the README. Seeds follow the per-build convention 2026070X–202607XX recorded in each config; the geo15 build uses 20260715 and the v10.3 exploratory stages use 20260716–20260721. Seed values are date-like *labels* chosen for readability, not timestamps, and carry no registration force. The v10.3 additions comprise `stage23_loso.py` through `stage28_powerfp.py`, one frozen config per stage (`run_config_loso23.json ... run_config_powerfp28.json`), and outputs prefixed `loso23_`, `taskpos24_`, `flat25_`, `count26_`, `gate27_`, `powerfp28_` in `tables/`; each plan was frozen before its computation. `RESULT_PROVENANCE.md` maps every number retained in the main text to its output file and verification status. Table S5 maps every main-text display to its generating script and source tables.

S10 Operational protocol and gate validation

The two-step gate, operationally. *Inputs:* the flagged integer partition and a clean reference partition (or the magnitude-matched uniform null below). *Step 1:* run the digit test that raised the concern; record its statistic and screen (e.g. Beber–Scacco χ^2_9 on records ≥ 1000). *Step 2:* compute the recording-grid coordinates. Defaults, as validated here: the modulus set $\{4, 8, 25, 49, 100, 47, 43\}$; modal grid g_0 , the largest g among 2, 4, 5, 10,

Table S5: Display provenance: every main-text figure and table, its generating script (`scripts_v10/`), and its source tables. “artifact hash” means the table is covered by the frozen `artifact/hashes/SHA256SUMS` or `claim_manifest.csv`; “ledger hash” means its SHA-256 is recorded in `RESULT_PROVENANCE.md`; `geo15` tables are exploratory outputs of `stage22_geo15.py` (seed 20260715).

display	script	source tables	hash coverage
Fig. 1	<code>fig1_cvap.py</code>	<code>vote14_panel.csv</code> , <code>vote14_grid_probe.csv</code> , <code>vote14_beber_scacco.csv</code>	artifact
Fig. 2	<code>fig2_geography.py</code>	<code>geo15_geography_channel.csv</code> , <code>geo15_controlled.csv</code>	geo15 (exploratory)
Fig. 3	<code>fig3_waterfall.py</code>	<code>derounding_effect.csv</code> , <code>deepstrip_effect.csv</code> , <code>hunt09_residual_rebaseline.csv</code> , <code>resonance_controls.csv</code> , <code>residue_structure.csv</code>	artifact + ledger
Fig. 4	<code>fig4_mechanism_power.py</code>	<code>hard10_degradation.csv</code> , <code>power13_quotient_spikein.csv</code>	artifact
Fig. 5	<code>fig5_acs.py</code>	<code>whisp12_acs.csv</code>	artifact
Fig. 6	<code>fig6_exact.py</code>	<code>gate_rungs.csv</code> , <code>hunt09_lowkeff.csv</code> , <code>family_profiles_v3.csv</code> (restricted to the 367 frozen families), <code>observational_corpus_v2.csv</code>	artifact
Table 1	—	<code>family_inventory.csv</code> (frozen corpus composition)	artifact
Table 2	—	<code>hunt09_residual_rebaseline.csv</code> and the frozen driver tables (<code>derounding_effect.csv</code> , <code>deepstrip_effect.csv</code> , <code>residue_structure.csv</code>)	artifact + ledger
Table 3	—	none (interpretive summary; no numeric claims)	—

20, 25, 50, 100, 200, 250, 500, 1000 with conformance ≥ 0.90 (else 1); magnitude-matched reference = uniform integers drawn to the series’ digit-length histogram (20 seeded draws; report excess = observed minus null-mean TV); z -scores against the reference draws’ spread when a formal scale is needed. *Multiplicity*: with 7 moduli and several partitions, treat single marginal excesses as screening only; the validated flag is TV excess ≥ 0.05 (an order of magnitude above the generic level ≈ 0.006) at any modulus, or $g_0 > 1$. *Stop/go*: flag present $\rightarrow$ report “the alarm is grid-compatible” and seek the documented recording rule (attribution requires the documentation); flag absent $\rightarrow$ “not attributable to a grid this instrument can see”—substantive investigation proceeds. The output is never a verdict on the data. A worked verification script for the CVAP row ships in the deposit README.

Worked CVAP report (every field of the main-text protocol). Unit of analysis: county tabulation (3,222 county-level rows in the special tabulation; the ACS/decennial county frame of main-text Figure 5 has 3,144; 3,171 rows pass the ≥ 1000 screen). Scored population: published `CVAP_EST`, 2018–2022 special tabulation. Representation: last digits (step 1); recording-grid coordinates (step 2). Baseline: last-digit uniformity; magnitude-matched uniform null. Known recording rule: the documented rounding maps estimates of 1–7 to 4 and rounds estimates ≥ 8 to the nearest five, so every analyzed value (all ≥ 1000) lies in the nearest-five regime and the five-grid statement is exact for the analyzed data (U.S. Census Bureau 2023). Result class: construction-attributed (signature + documentation). Detector outcome: $\chi^2_9 = 12,684$, p below reporting precision. Gate outcome: $g_0 = 5$; TV excess +0.765 at modulus 25, +0.729 at 100. Compatible alternative mechanisms: none required beyond the documented rule. Licensed conclusion: the alarm is attributable to the documented five-grid. Not licensed: any statement about the accuracy or integrity of the underlying tabulation.

Adversarial co-occurrence example (v10.3, seed 20260720). Fabricated *and* gridded: values drawn from a smooth lognormal generator (“fabricated”: no real process behind them), then published on a five-grid, fire the last-digit test at $\chi_9^2 \approx 11,800$ and show $g_0 = 5$ with TV excess +0.76 at modulus 25—the gate correctly reports grid-compatible, because the *alarm* is caused by the grid. The example operationalizes the warning: the gate attributes the alarm, not the data; a fabricated series published on a grid passes the gate exactly like an authentic one.

Ground-truth validation library (v10.3, seed 20260720). Recipe frozen in `run_config_gate27.json`; output `gate27_library.csv`. Sensitivity is computed over *seven* grid-truth series: the three documented real-data series (CVAP, CES, ACS margin-of-error) plus four constructed ones (the fabricated-and-gridded synthetic of the preceding paragraph, the binary-aligned allocation demo, and the 47- and 43-tick re-quantizations). Coordinate sets: the corpus instrument’s eleven coordinates use five residue moduli (100, 25, 49, 47, 43); the operational default here extends the set to seven (adding 4 and 8 for binary grids). Documented-grid series: CVAP county (flagged, $g_0=5$); CES survey employment, hundreds-grid (flagged, $g_0=100$, TV excess +0.92); ACS margin-of-error (**missed**: its partial rounding sits below the 0.05 coarse threshold and 0.90 conformance bar—this is the case that requires the CI-based grid-mass comparison of the main text); binary-aligned allocations (flagged, $g_0=4$); constructed 47- and 43-ticks (flagged at their own moduli with $g_0=1$ —the non-decimal capability). Non-grid series: county returns 2020/2024, decennial-style county population counts, additive synthetic sums, uniform null (all clean); NC precinct returns (**false-flagged** at modulus 8, excess +0.116)—a small-magnitude series in exactly the failure regime named by the flatness stress of §S2. Sensitivity 6/7 (the one miss is a documented real series); specificity 5/6; both misses mechanistically explained. The library doubles as the validation table for the main-text protocol; its scope is series already cached by the project—no new acquisition.

References

- Richard Arratia, A. D. Barbour, and Simon Tavaré. Limits of logarithmic combinatorial structures. *The Annals of Probability*, 28(4):1620–1644, 2000. doi: 10.1214/aop/1019160500.
- Patrick Billingsley. On the distribution of large prime divisors. *Periodica Mathematica Hungarica*, 2(1–4):283–289, 1972. doi: 10.1007/BF02018667.
- Karl Dickman. On the frequency of numbers containing prime factors of a certain relative magnitude. *Arkiv för Matematik, Astronomi och Fysik*, 22A(10):1–14, 1930.
- Saulius Gražulis et al. [dataset] crystallography open database (COD). <https://www.crystallography.net/cod/>, 2024. CC0; per-structure Wyckoff multiplicities. Accessed 2026.
- J. F. C. Kingman. Random discrete distributions. *Journal of the Royal Statistical Society, Series B*, 37(1):1–22, 1975. doi: 10.1111/j.2517-6161.1975.tb01024.x.

- Ernst Eduard Kummer. Über die Ergänzungssätze zu den allgemeinen Reciprocitätsgesetzen. *Journal für die reine und angewandte Mathematik*, 44:93–146, 1852.
- Adrien-Marie Legendre. *Essai sur la Théorie des Nombres*. Courcier, Paris, 2nd edition, 1808.
- Jure Leskovec and Andrej Krevl. [dataset] SNAP datasets: Stanford large network dataset collection. <https://snap.stanford.edu/data>, 2014. Collaboration graphs used for automorphism orders. Accessed 2026.
- NIST. [dataset] NIST atomic spectra database. <https://www.nist.gov/pml/atomic-spectra-database>, 2024. Level degeneracies $2J + 1$. Accessed 2026.
- Ákos Seress. *Permutation Group Algorithms*. Cambridge University Press, Cambridge, 2003.
- U.S. Census Bureau. Citizen Voting Age Population (CVAP) special tabulation from the 2018–2022 5-year American Community Survey: Technical documentation. https://www2.census.gov/programs-surveys/decennial/rdo/technical-documentation/special-tabulation/CVAP_2018-2022_ACS_documentation_v1.pdf, 2023. States the rounding rule for published CVAP_EST values. Accessed 2026.